
Annex 3 — Output Markets and Pricing

Purpose: Justifies the unit prices used in the financial model for each of the four marketable output streams produced by the Urban Rig: pyrolysis oil (three fractions), biochar/carbide (five market tiers), recovered metals, and carbon credits. Each price is benchmarked against current European market data with inline sources.

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3.1 Pyrolysis Oil — Three Fractions

3.1.1 The Fractional Separation Advantage

Unlike most pyrolysis technologies that produce a single crude oil stream requiring downstream distillation, the Urban Rig URC-2000 collects oil vapours at three distinct temperature points as waste moves through the 7-zone reactor. Each fraction is condensed independently via separate solenoid valves and condensation paths (Urban Rig Recycling Solutions ENG, p. 15):

Zone	Temperature	Product	Carbon Range	Share of Liquid Oil	Daily Volume (URC-2000)
1	~150°C	Light oil (naphtha range)	C ₅ -C ₈	10%	7,803 L
2	~300°C	Gas (diesel) oil	C ₈ -C ₁₆	80%	62,424 L
3	~450°C	Heavy oil (waxes/paraffins)	C ₁₆ +	10%	7,803 L

Source: Urban Rig Recycling Solutions ENG, pp. 15–16. Volume data from URC-2000 output table at 90% resin content, 200 T/day volumetric.

This built-in fractional distillation allows each fraction to be sold at its grade-specific market price rather than accepting the discount associated with crude, undifferentiated pyrolysis oil.

3.1.2 Oil Quality — Measured Data

NKKK Fuel Certification (Nippon Kaiji Kentei Kyokai — Japan Marine Survey Association, JIS standards):

Parameter	Result	Comparison
Calorific value	43.27 MJ/kg	Fossil diesel: 42.6 MJ/kg
Flashpoint	$\leq 40^{\circ}\text{C}$	Contains light naphtha fractions
Sulfur	0.014% (140 ppm)	Below IMO 2020 (0.5%); above Euro V diesel (10 ppm)
Chlorine	0.0001% (1 ppm)	Well below problematic levels
Pour point	$\leq -20^{\circ}\text{C}$ <i>Goodcold — flowproperties</i> <i>Viscosity(50°C)</i> 1.11 mm ² /s	Light, low-viscosity product
Ash	0.005%	Very clean

Source: NKKK certified test report (JIS standards). Note: sample may be light fraction only — verification with UR/OneWorld recommended.

GC-MS Analysis (Osaka Institute of Industrial Science and Technology) confirmed the volatile fraction is dominated by middle-distillate hydrocarbons: dimethylheptane (C₉), undecene (C₁₁), and trimethylnonane (C₁₂) — the C₈–C₁₆ range directly usable as naphtha-substitute feedstock for steam crackers (Technical Findings Log §12).

3.1.3 Market Benchmarks (2024–2025)

Fossil equivalents:

Product	Market	Price (EUR/T)	Source
Naphtha (NWE)	ARA spot	~€485–660	ChemAnalyst Q4 2025; Intratec Netherlands May 2025
ICE Low Sulphur Gasoil	NWE futures	~€820	ICE LS Gasoil futures 2025

Product	Market	Price (EUR/T)	Source
Heavy Fuel Oil (380 cst)	Rotterdam bunker	~€415	S&P Global Platts Feb 2025

Recycled pyrolysis oil (ICIS Index):

ICIS launched the first pyrolysis oil pricing index in 2023, covering European spot transactions. Key 2024–2025 data:

Grade	Price Indicator	Source
Tire-derived pyrolysis oil (baseline)	Base reference	ICIS weekly report
Plastic-derived PPO, non-upgraded, ex-works NWE	+€681/T premium over tire-derived (2024 avg)	ICIS via PlasticsToday
Plastic-derived naphtha- substitute (cracker grade)	+€1,114/T premium over tire-derived (2024 avg)	ICIS via Tire Technology International
Netherlands PPO spot Q1 2025	~\$729/T ≈ €670/T	IMARC Pyrolysis Oil Price Trends 2025

Recycled content premium driver: EU regulations are creating sustained demand for recycled-content petrochemical feedstock. The PPWR (Packaging and Packaging Waste Regulation) mandates recycled content targets for plastic packaging — 30% by 2030, 65% by 2040 (EU PPWR, adopted 2024). The EU ETS makes fossil feedstock relatively more expensive. ISCC PLUS certification enables mass balance accounting, allowing refineries to attribute recycled content to downstream products. BASF ChemCycling, TotalEnergies, and Esso/ExxonMobil all have active offtake programmes for ISCC-certified pyrolysis oil (BASF press release March 2025; TotalEnergies sustainability reports).

3.1.4 Oil Pricing Model — Three Scenarios

Price per fraction (EUR/T):

Fraction	Share	Low	Medium	High	Rationale
Light oil (C ₅ -C ₈)	10%	450	600	800	Low: discount for quality uncertainty. Medium: naphtha parity. High: + recycled premium
Gas (diesel) oil (C ₈ -C ₁₆)	80%	500	670	950	Low: crude PPO floor. Medium: ICIS spot NWE 2025. High: naphtha-sub with ISCC premium
Heavy oil (C ₁₆ +)	10%	350	450	550	Low: HFO discount. Medium: marine fuel/industrial. High: hydrocracker feed + green premium

Weighted average oil price:

Scenario	Weighted EUR/T	EUR/L (at 0.85 kg/L)	Basis
Low (conservative)	€480/T	€0.41/L	No green premium. All sold as undifferentiated crude. Floor price

Scenario	Weighted EUR/T	EUR/L (at 0.85 kg/L)	Basis
Medium (base case)	€641/T	€0.55/L	Current ICIS spot. Fractions sold at grade-specific prices. Recommended for business plan
High (optimistic)	€895/T	€0.76/L	Full recycled premium. ISCC-certified. Long-term offtake with cracker operators

Calculation: Weighted price = $\sum$ (fraction share $\times$ fraction price). Example Medium: $0.10 \times 600 + 0.80 \times 670 + 0.10 \times 450 = \text{€}641/\text{T}$.

Note: The original financial model used a hardcoded oil price of €0.41/L (€482/T) — corresponding to the Low scenario (floor price, zero green premium, undifferentiated crude). The Medium scenario (€641/T) is the appropriate base case given that 80% of UR output is middle distillate and the machine already separates fractions without downstream processing.

3.1.5 Target Buyers

Fraction	Target buyers	Sales channel
Light oil (naphtha)	Naphtha cracker operators (BASF, TotalEnergies, Esso/ExxonMobil)	ISCC PLUS certified offtake contracts
Gas oil (diesel)	Diesel/kerosene blenders, refinery co-processors	Spot + term contracts via ICIS-referenced pricing
Heavy oil	Marine fuel traders, hydrocracker feedstock buyers	Industrial spot or burn on-site for reactor heating

3.1.6 Recommendations

1. **Obtain indicative offtake pricing** from at least two potential buyers (e.g., TotalEnergies Gonfreville, Esso Fos-sur-Mer) for the diesel fraction specifically. This converts the Medium scenario from hypothesis to confirmed market price.
2. **Plan ISCC PLUS certification from Day 1** — it unlocks the full recycled premium that justifies the Medium and High scenarios.
3. **Design separate storage tanks** for the three fractions to preserve the price differentiation advantage.
4. **Run the financial model at all three scenarios** (Low/Medium/High) to show investors the range of outcomes.

3.2 Biochar / Carbide

3.2.1 Product Characteristics

The Urban Rig produces 480.8 kg of combustible char per 1,000 kg of OMR (57.5% of reactor input after inert removal — see Annex 5 §11.4). In the V2 model, all char is sold as fuel; none is diverted for reactor heating (which is supplied by purchased heating oil). Total char available for sale is approximately **67 T/day (~22,155 T/year)**.

Char characteristics depend on source material:

Source	Share of combustible char	Estimated CV (MJ/kg)	Carbon content	Data quality
Organic/composts char	94% (452.2 / 480.8 kg)	15	~40-50%	HYPOTHESIS — literature (Fazil et al., <i>Energies</i> , 2021; Ronsse et al., 2020)
Plastic char	6% (28.6 / 480.8 kg)	25	~60-80%	HYPOTHESIS — consistent with carbon black (~32 MJ/kg). One UR measurement: urethane char at 32.09 MJ/kg (Isumi Tech, JIS M 8814)
Weighted average	100%	15.6	~42-52%	

3.2.2 Market Tiers

The surplus char can be sold into several markets depending on quality, certification, and form factor (loose vs compressed pellets/briquettes):

Tier	Market	Price (EUR/T)	Basis	Annual Revenue (16,088 T)	Total Benefit (incl. €1M diesel saved)
T1	Waste-to-energy fuel (RDF substitute)	50	Sold as-is to cement kilns or WtE plants	€804k	€1.8M
T2	Industrial fuel pellets	140	Compressed into pellets, whole-sale fuel market	€1.6M	€2.6M
T3	Charcoal briquettes	200	Compressed briquettes for heating/BBQ market	€3.2M	€4.2M
T4	Biochar for agriculture	350	Soil amendment, requires EBC certification	€5.6M	€6.6M

Tier	Price Market (EUR/T)	Annual Revenue Basis (16,088 T)	Total Benefit (incl. €1M diesel saved)
T5	Biochar 350 + + carbon credit bon cred- its	Physical sale + CORC/EBC carbon removal certifi- cate	€5.6M + €3.2M €9.8M

Sources: Tier 1-3 pricing from European solid fuel market benchmarks. Tier 4 from European Biochar Certificate (EBC) marketplace. Tier 5 carbon credit pricing from Puro.earth CORC Carbon Removal Price Indexes (2025), CDR.fyi Biochar Carbon Removal Market Snapshot (2025), carboncredits.com biochar market analysis (2025).

Tier 2 (€100/T) is the recommended base case for the financial model — conservative, does not require certification, and targets an established market for industrial fuel pellets. The current Excel model already uses €100/T (UR-Financial-Model-General.xlsx, Revenue sheet).

3.2.3 Carbon Credit Potential (Tier 5)

MSW-derived char has ~40–50% carbon content. Carbon sequestration potential:

Parameter	Value
Carbon content of char	~42–52% (weighted estimate)
CO ₂ equivalent per tonne of char	~1.5 T CO ₂ (conservative)
Annual surplus char	16,088 T/year
Annual CO ₂ sequestered	~24,132 T CO ₂ /year
CORC price (2025)	~€133/T CO ₂
EU ETS price (conservative)	~€50/T CO ₂
Revenue at CORC price	~€3.2M/year
Revenue at EU ETS price	~€1.2M/year

Sources: CORC biochar carbon removal index from Puro.earth (2025). EU ETS price from ICE EUA futures (2025).

This revenue is **additional** to the physical sale of char (Tier 4 or 5 combined). However, biochar carbon credits require EBC or Puro.earth certification, soil application (not combustion), and chain-of-custody documentation. This pathway is best suited for Phase 2 after validating char quality.

3.2.4 Pelletisation CAPEX

Compressing char into pellets or briquettes requires a pelletising unit. Indicative cost for an industrial pelletiser capable of 50 T/day throughput: €200k–500k. This is a minor addition to the €50M+ CAPEX and is not separately modelled; it would appear in the “Site infrastructure” line of the CAPEX sheet (Annex 5 §13.6).

3.3 Recovered Metals

3.3.1 Recovery Mechanism

Metals in the waste stream — ferrous (steel) and non-ferrous (aluminium, copper) — are recovered in **non-oxidised form** because the anoxic reactor environment (superheated steam at 600°C, zero oxygen) prevents oxidation. This contrasts with incineration where metals are oxidised and must be recovered from toxic bottom ash at reduced quality and value (Urban Rig Recycling Solutions ENG, p. 13; Eurostat waste incineration guidelines).

3.3.2 Metal Content in MSW

Per ADEME MODECOM 2017, metals represent approximately 3% of OMR (30 kg per 1,000 kg). An additional 3 kg of metal fragments is recovered from the fine elements fraction (Annex 5 §2). Total recoverable metals: approximately **33 kg per tonne of OMR**, split roughly 70% ferrous / 30% non-ferrous (ADEME 2017 characterisation).

3.3.3 Pricing

Metal type	Share	Price (EUR/T)	Source	Revenue per 1,000 kg OMR
Ferrous (steel scrap)	70% (~23 kg)	250–350	European steel scrap index (LKAB Minerals, S&P Global Platts, 2024– 2025)	€5.75–8.05
Non-ferrous (aluminium, copper mix)	30% (~10 kg)	1,200–2,500	LME alu- minium ~€2,200/T; LME copper ~€8,500/T; blended non- ferrous ~€1,800/T	€12.00–25.00
Total metals	33 kg	—	—	€17.75–33.05

*Scaling to 140 T/day × 330 days = 46,200 T/year OMR: approximately **€0.8M–1.5M/year** in metal revenue.*

The financial model uses **€0.7M/year** (UR-Financial-Model-General.xlsx, Revenue sheet) — at the conservative end of the range, reflecting sorting/cleaning losses and market variability.

3.3.4 Non-Oxidised Quality Premium

The non-oxidised recovery condition is a meaningful advantage. Metals recovered from incineration ash are typically oxidised, contaminated with slag, and sell at a discount of 30–50% versus clean scrap (Eurostat, CEWEP incineration benchmarks). Urban Rig metals, recovered in their original metallic form, command full scrap market prices without quality discounts.

3.4 Carbon Credits

3.4.1 Carbon Credit Mechanism

Carbon credits are generated by the difference between Urban Rig CO₂ emissions and the counterfactual emissions from the waste treatment pathway it displaces (typically incineration). Two distinct credit sources exist:

- 1. **CO₂ avoided** — less fossil carbon emitted than incineration of the same waste
- 2. **CO₂ sequestered** — carbon permanently stored in biochar applied to soil (separate from Tier 5 biochar credits in §3.2.3)

3.4.2 CO₂ Avoided vs Incineration Baseline

Emission source	Incineration (counterfactual)	Urban Rig	Difference
Combustion CO ₂ (plastic fraction)	~1.0 T CO ₂ /T waste	~0 (no combustion)	\$-1.0T BiogenicCO_2\$ (organic fraction)

Sources: ADEME incineration emission factors (2022); IPCC waste-sector guidelines; financial model carbon calculations.

At 46,200 T/year throughput: approximately **46,200-69,300 T CO₂/year** avoided.

3.4.3 Carbon Credit Pricing

Market	Price (EUR/T CO ₂)	Source	Annual revenue estimate
EU ETS (compliance market)	~€50-80	ICE EUA futures (2024-2025)	€2.3M-5.5M

Market	Price (EUR/T CO ₂)	Source	Annual revenue estimate
Voluntary carbon market (VCM)	~€15-40	Ecosystem Market-place, MSCI Carbon Markets (2024)	€0.7M-2.8M
Financial model assumption	€50	Conservative (EU ETS floor)	€2.5M

Source: UR-Financial-Model-General.xlsx, Revenue sheet — uses €50/T CO₂ and approximately 50,000 T CO₂/year net reduction.

Eligibility note: Carbon credits require formal methodology registration (e.g., Gold Standard, Verra VCS, or EU ETS inclusion). The UR process would need to demonstrate additionality and secure third-party verification. This is a Phase 2 revenue stream — conservatively included at €50/T but requiring regulatory and certification work.

3.5 Revenue Summary — Financial Model Reference

At the base case (URC-2000, 200 T/day, 30% plastic content, Year 3 steady state, Medium oil scenario):

Revenue Stream	Annual Estimate	% of Total	Pricing Basis
Gate fees (waste intake)	€5.3M	31%	€80-180/T tipping fee range (ADEME 2022); model uses blended €114/T
Fuel oil sales (3 fractions)	€7.1M	41%	Medium scenario €641/T weighted (ICIS 2025; see §3.1.4)

Revenue Stream	Annual Estimate	% of Total	Pricing Basis
Char/biochar sales (surplus)	€1.7M	10%	Tier 2 at €100/T (fuel pellets; see §3.2.2)
Recovered metals	€0.7M	4%	Conservative blend of ferrous + non-ferrous (see §3.3.3)
Carbon credits	€2.5M	14%	€50/T CO ₂ × ~50,000 T/year (see §3.4.3)
TOTAL	€17.3M	100%	

Source: UR-Financial-Model-General.xlsx, Revenue sheet (Year 3 steady state). 815 validated formulas, 0 circular references, 0 errors.

3.5.1 Revenue Sensitivity to Oil Price

The oil price is the single most impactful assumption after machine cost:

Oil Scenario	Oil Revenue	Total Revenue	EBITDA	Change vs Medium
Low (€480/T)	~€5.3M	~€15.6M	~€10.2M	—€1.7M
Medium (€641/T)	~€7.1M	~€17.3M	~€11.9M	Base case
High (€895/T)	~€9.9M	~€20.1M	~€14.7M	+€2.8M

Moving from Low to Medium adds ~€1.7M/year; from Medium to High adds ~€2.8M/year.

3.6 Data Quality Summary

Output	Price Used	Classification	Confidence
Oil — fraction split (10/80/10)	From UR spec	FACT	MEDIUM — at 90% resin; actual split may vary with MSW feedstock
Oil — Medium price €641/T	ICIS benchmark	FACT (market data)	MEDIUM-HIGH — 2024–2025 spot; no UR offtake contracts yet
Oil — NKKK quality (43.27 MJ/kg)	Certified lab	FACT	HIGH — note: may be light fraction only
Biochar — T2 at €100/T	Market benchmark	REASONABLE	MEDIUM — established fuel pellet market
Biochar — char CV 15.6 MJ/kg	Literature	HYPOTHESIS	MEDIUM — not measured on UR MSW char
Metals — €0.7M/year	Market indices	REASONABLE	MEDIUM — depends on sorting efficiency
Carbon credits — €50/T CO ₂	EU ETS floor	REASONABLE	MEDIUM — requires methodology + verification
Gate fees — €114/T blended	ADEME range	REASONABLE	HIGH — well within French tipping fee corridor

3.7 References

Urban Rig Technical Sources

1. **Urban Rig Recycling Solutions ENG** — OneWorld Corporation product brochure (English edition). Cited for: fractional oil separation design (pp. 15–16), metal recovery in anoxic conditions (p. 13).
2. **NKKK (Nippon Kaiji Kentei Kyokai)** — Recycled fuel test report, JIS standards. Cited for: oil quality certification — 43.27 MJ/kg CV, 0.014% sulphur, 1

ppm chlorine (§3.1.2).

3. **Osaka Institute of Industrial Science and Technology** — GC-MS analysis of volatile oil fraction. Cited for: compound identification (dimethylheptane, undecene, trimethylnonane) in Technical Findings Log §12 (§3.1.2).

Market Data and Pricing Sources

4. **ICIS** — Weekly pyrolysis oil pricing index (launched 2023). Cited for: European spot PPO pricing, tire-derived baseline, plastic-derived premiums (§3.1.3, §3.1.4).
5. **ChemAnalyst** — Naphtha NWE pricing Q4 2025. Cited for: naphtha benchmark €485–660/T (§3.1.3).
6. **Intratec** — Netherlands naphtha pricing May 2025. Cited for: naphtha benchmark (§3.1.3).
7. **ICE (Intercontinental Exchange)** — Low Sulphur Gasoil NWE futures 2025; EUA futures 2024–2025. Cited for: gasoil ~€820/T (§3.1.3), EU ETS carbon price €50–80/T (§3.4.3).
8. **S&P Global Platts** — Rotterdam bunker HFO 380 cst (Feb 2025); European steel scrap index. Cited for: HFO ~€415/T (§3.1.3), ferrous scrap €250–350/T (§3.3.3).
9. **IMARC** — Pyrolysis Oil Price Trends 2025. Cited for: Netherlands PPO spot ~\$729/T (§3.1.3).
10. **LME (London Metal Exchange)** — Aluminium and copper spot prices 2024–2025. Cited for: non-ferrous metal pricing (§3.3.3).

Regulatory and Industry Sources

11. **EU PPWR (Packaging and Packaging Waste Regulation)** — Adopted 2024. Cited for: recycled content mandates — 30% by 2030, 65% by 2040 (§3.1.3).
12. **BASF** — ChemCycling press release (March 2025). Cited for: recycled feedstock offtake programmes (§3.1.3).
13. **TotalEnergies** — Sustainability reports. Cited for: ISCC-certified pyrolysis oil offtake (§3.1.3).
14. **ISCC PLUS** — International Sustainability and Carbon Certification. Cited for: mass balance certification enabling recycled content claims (§3.1.3).

Biochar and Carbon Credit Sources

15. **European Biochar Certificate (EBC)** — Marketplace pricing. Cited for: Tier 4 biochar agricultural pricing (§3.2.2).
16. **Puro.earth** — CORC Carbon Removal Price Indexes (2025). Cited for: biochar carbon credit pricing ~€133/T CO₂ (§3.2.3).
17. **CDR.fyi** — Biochar Carbon Removal Market Snapshot (2025). Cited for: carbon removal market benchmarks (§3.2.2).
18. **carboncredits.com** — Biochar market analysis (2025). Cited for: carbon credit market context (§3.2.2).

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19. **Ecosystem Marketplace** — Voluntary carbon market data (2024). Cited for: VCM pricing €15–40/T (§3.4.3).
 20. **MSCI Carbon Markets** — Carbon market analysis (2024). Cited for: voluntary market benchmarks (§3.4.3).

Institutional Sources

21. **ADEME** — Incineration emission factors (2022); tipping fee ranges. Cited for: CO₂ avoided calculations (§3.4.2), gate fee corridor (§3.5).
22. **IPCC** — Waste-sector greenhouse gas guidelines. Cited for: emission factor methodology (§3.4.2).
23. **Eurostat** — Waste incineration guidelines; CEWEP benchmarks. Cited for: metal quality discount in incineration (§3.3.4).

Internal Model References

24. **UR-Financial-Model-General.xlsx** — Working financial model (CAPEX, Revenue, P&L sheets). Cited for: revenue assumptions, oil price baseline, steady-state projections (§3.5).

End of Annex 3